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Entity and relation extractions for threat intelligence knowledge graphs

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ABSTRACT

Advanced persistent threats (APTs) represent a complex challenge in cybersecurity as they infiltrate networks stealthily to conduct espionage, steal data, and maintain a long-term presence. To combat these threats, security professionals increasingly rely on cyber knowledge graphs (CKGs), which provide scalable solutions to analyze and structure vast amounts of cyber threat intelligence (CTI) from diverse sources in real-time, enabling the automation of proactive security measures. Developing CKGs requires extracting entity and their relationships from unstructured CTI reports. However, existing approaches face significant limitations, such as difficulties with the nuances of cybersecurity language, diverse threat terminologies, and high rates of error propagation, resulting in low accuracy and poor generalizability. This paper introduces a novel Threat Intelligence Knowledge Graph (TiKG) pipeline designed to address these challenges. The TiKG framework leverages SecureBERT, a domain-specific transformer-based model optimized for cybersecurity, and integrates it with an attention-based BiLSTM to capture the context and nuances of security texts, reducing error propagation and improving extraction accuracy. Additionally, the pipeline incorporates a domain-specific ontology and inference model to ensure precise relation mapping in relation extraction. Using three large-scale TI open-source datasets (DNRTI, STUCCO, and CYNER) and a curated CTI dataset, extensive evaluations demonstrate the effectiveness of our framework, showing significant improvements over existing methods in detecting and linking cyber threats. These contributions provide a robust platform for security professionals to analyze and predict potential attacks, develop effective defenses, and enhance the strategic capabilities of cybersecurity operations.

Problems of relaibility and error propagation du
entity relation extraction

1. Introduction

Secutity rely on CTI (unstructured repor
to improve cybersetuciry
strategies. The extraction must be
precise

Advanced persistent threats (APTs) are sophisticated cyberattacks aimed at gaining unauthorized, long-term access to computer networks for strategic objectives like espionage and data theft. These threats pose significant risks to national security, corporate intellectual property, and personal privacy, necessitating targeted and continuous defensive efforts. Security experts rely on past cyber threat intelligence (CTI) reports — drawn from sources such as news articles, technical reports, and security logs — to combat APTs. These unstructured reports offer insights into threat actors, vulnerabilities, attack vectors, and attackers' tactics, techniques, procedures (TTPs), and tools. By extracting actionable intelligence and relationships from these reports and representing them in cyber knowledge graphs (CKGs), security professionals can develop robust defensive strategies. CKGs enable real-time analysis, integrate diverse data sources, and enhance the understanding of threat landscapes, helping to design proactive defenses and predict potential attack paths (Injy Sarhan, 2021; Wang and Liu, 2023; Pingle et al., 2019). As a result, CKGs have become essential tools in cybersecurity defense, particularly against emerging APTs.

The growing importance of CKGs in both academia and industry is driven by the need to structure data for improved security operations (Ahmed et al., 2024; Guo et al., 2021; Li et al., 2022b; Injy Sarhan, 2021). CKGs involve two primary tasks: extracting entities from unstructured texts via named entity recognition (NER) and identifying relationships between these entities through relation extraction (RE). While joint extraction combines NER and RE in a single NLP model, pipeline extraction separates these tasks.

Extracting entities and relationships from unstructured text poses significant challenges, including language ambiguity, diverse threat terminologies, complex cyber threat narratives, and the number of entities and relations in the dataset. These factors contribute to error propagation and limit the accuracy, scalability, and generalizability of existing pipeline approaches (Ahmed et al., 2024; Peng et al., 2022). As a result, recent studies have adopted joint extraction techniques to improve entity and relation extraction. Pipeline approaches, such as those by Jo et al. (2022), have achieved competitive results using private datasets with fewer entities and relations.

This paper introduces a novel pipeline architecture with enhanced error control for accurately extracting entities and relations from CTI

Making a CKG (cyber knowledge graph) requires 2 steps :

--> Named entity recognition

-->relationships between these entities extraction

But source text is unstructured so it generates problems : langage
ambiguity, diverse terminology, identification of the relations, ...

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Reminder : RE = Relation Extractio
NER = Named Entity Recognition

reports to build robust CKGs. As shown in Fig. 1, the pipeline couples the RE inference model with a domain-specific ontology, ensuring that every extracted pair of entities is connected with the correct relation. To minimize the likelihood of error propagation and improve entity recognition accuracy, it employs a hybrid NER model leveraging SecureBERT embeddings, which captures cybersecurity-specific terminology, jargon, and nuances often overlooked by the general-purpose models (Aghaei et al., 2023) used in existing works. This novel approach offers several advantages, delivering strong performance even with complex datasets containing numerous entities and relations. The error control in the RE task enables the construction of robust knowledge graphs, accurate attack forecasting, and the deployment of protective strategies against real threats. Developing this level of error control is challenging in joint extraction models that combine NER and RE into a single process. Our proposed pipeline serves as a novel candidate for comparing pipeline and joint approaches, a task beyond the scope of this work. Such a comparison would require a comprehensive study involving various parameters like feature confusion, model design, system settings, and data complexity (Yan et al., 2022).

To validate our model, enhance scalability, and ensure generalizability and reproducibility, this study utilizes three large-scale, open-source NER datasets. The DNRTI dataset (Wang et al., 2020b) includes 13 entity categories across 6592 sentences describing APT groups. Despite its potential, DNRTI's effectiveness in threat detection and knowledge graph construction remains underexplored. We predefined 15 unique relations and applied advanced algorithms to detect threat-related entities and their interconnections within unstructured security reports. The two additional NER datasets used for validation are: (i) STUCCO (Bridges et al., 2014), featuring 15 entity types and 15,192 sentences from CVE and NVD descriptions; and (ii) CyNER (Alam et al., 2022), encompassing 5 entity classes across 4372 sentences.

The key contributions of this paper are summarized as follows:

- We introduced a novel end-to-end extraction framework for Threat Intelligence Knowledge Graphs (TiKG) that accurately extracts entities and relationships from unstructured CTI reports. Unlike existing pipelines, it combines an advanced inference model with expert knowledge encoded in an ontology to eliminate relation classification errors. Additionally, it integrates a context-aware model using SecureBERT embeddings to understand CTI nuances, minimizing error propagation and enhancing entity recognition accuracy. It stores extracted entity-relation triples using Neo4j, enabling complex queries that allow security analysts to correlate large volumes of APT reports and extract timely insights. This design enables the framework to handle complex datasets with numerous entities and relations, ensuring high-quality knowledge graphs.
- We conducted a comprehensive ablation study on our framework using three large-scale threat intelligence datasets (DNRTI, CyNER, and STUCCO) to assess its effectiveness in constructing CKGs. We compared the performance of our models against recent state-of-the-art (SOTA) models in the field. To demonstrate the applicability and quality of the resulting knowledge graph, we designed a use case scenario addressing a real-world problem, solving it with search queries to directly benefit security practitioners.
- The curated CTI reports, along with the public NER datasets used for validation, are made available on our GitHub[1] to foster research in the field and support the reproduction of this work.

The remainder of this paper is organized as follows: Section 2 discusses existing works and highlights identified gaps. Section 3 presents our proposed extraction framework and conducts an ablation analysis of each component. Section 4 summarizes the findings and contributions, and lays the foundation for future work.

[1] https://github.com/imouiche/Threat-Intelligence-Knowledge-Graphs.

2. Related works

This section reviews the existing literature focusing on entity recognition and relation extraction for the construction of CKGs and their automated analysis.

2.1. Entity recognition

Before: Rule base or dictionary based th
Now :deep learning algorithms sutch as Bidirectional long short-term memory (BILSTM)

Entity extraction in threat intelligence is a specific task within information extraction aimed at identifying and classifying cybersecurity-related entities. These entities include malware, threat actors, indicators of compromise (IoC), security tools, vulnerabilities, and others, found within unstructured security reports. Initially, research on entity extraction predominantly employed rule-based or dictionary-based methods, where experts crafted numerous rule templates and utilized string matching to extract entities (Liu and Li, 2014). Although effective for recognizing fixed patterns, this approach often lacks generalization capabilities and struggles to adapt to the evolving nature of data.

Statistical machine learning offers improved outcomes for cybersecurity entity extraction tasks without the need for manually defining rule templates, thus enhancing portability. Mulwad et al. (2011) utilized a support vector machine (SVM) to classify attack means and outcomes, while Bridges et al. (2017) applied the maximum entropy model (MEM) across various security corpora to mitigate model overfitting. These methods are more robust than rule-based approaches. However, they still require manual extraction of text features, making it challenging to achieve satisfactory results on larger datasets.

The advent of deep learning has seen neural networks make significant strides, consistently delivering outstanding results in entity extraction tasks. An example is Dionísio et al. (2019), who initially employed a convolutional neural network (CNN) to identify whether tweets contained security information pertinent to IT infrastructure properties. Subsequently, they used a bidirectional long short-term memory (BiLSTM) network to extract named entities, thereby generating security alerts. Wang et al. (2020b, 2022) utilized both BiLSTM and BERT transformer-based models to demonstrate the effectiveness of their manually constructed datasets in recognizing named entities for threat intelligence. Similarly, Varghese et al. (2023) used a state-of-the-art NLP-based transformer model to accurately extract actionable threat intelligence information such as HackerIDs, tools, software, organizations, and other entities from dark web forums.

2.2. Relation extraction and knowledge graph

Relation extraction, a subtask of information extraction, involves identifying the links between subject and object entities. Similar to entity extraction, the main approaches to relation extraction can be categorized into three types: (i) early rule-based approaches (McDonald et al., 2005; Iria, 2005) relying on predefined rules created by experts, which can be effective but often lack flexibility and scalability; (ii) traditional machine learning-based approaches (Culotta and Sorensen, 2004; Jiang and Zhai, 2007) which require manual feature extraction and are more adaptable than rule-based approaches but still depend on domain expertise to define features effectively; and (iii) deep learning-based approaches (Zeng et al., 2014; Wei et al., 2020; Bekoulis et al., 2018; Zheng et al., 2017). In recent years, deep learning has become the focus of cutting-edge research in relation extraction. These models leverage neural networks that automatically learn to extract features from raw data, eliminating the need for manual feature extraction and extensive domain knowledge. This shift has enhanced the performance of relation extraction systems, enabling them to generalize better across different datasets and domains.

Two primary methods are employed to extract threat entities and their relationships from security reports for constructing CKGs: joint extraction and pipeline extraction techniques. In this context, the extracted threat entities and relations are commonly referred to as entity-relation triples or relation triples and are represented in the format

Trois techniques d'extraction de relations : early rule based qui dépend de règles crées par des experts mais qui n'est pas flexible ni
Le ML classique qui demande une extraction manuelle mais qui est plus adaptable mais qui dépend du domaine d'expertise pour ê
Le deep learning qui extrait automatiquement les features de data brutes ce qui enlève le besoin d'extraction manuelle.
Deux méthodes principales : joint extraction et pipeline extraction

Table 1
Relevant studies on entities and relations extraction. JE: Joint Extraction; PE: Pipeline Extraction; Att: Attention; KG: Knowledge Graph; GA: Graph Analysis.

Ref.	Technique	Dataset	Entities	Relations	JE or PE	KG	GA
Wang et al. (2020c)	C-GCN	Public	⊠	☑	JE	⊠	⊠
Miwa and Bansal (2016)	LSTM-RNNs	Public	☑	☑	JE	⊠	⊠
Bekoulis et al. (2018)	CRF Multi-headed selection	Public	☑	☑	JE	⊠	⊠
Yuan et al. (2020)	BiLSTM + Att	Public	☑	☑	JE	⊠	⊠
Zheng et al. (2017)	BiLSTM	Public	☑	☑	JE	⊠	⊠
Guo et al. (2021)	BERT-BiGRU-CRF	Custom	☑	☑	JE	⊠	⊠
Zuo et al. (2022)	BERT-BiLSTM-CRF + Att	Custom	☑	☑	JE	⊠	⊠
Ahmed et al. (2024)	RoBERTa-BiGRU-CRF	Custom	☑	☑	JE	⊠	⊠
Zhao et al. (2020)	BiLSTM-CRF	Custom	☑	☑	PE	⊠	⊠
Gasmi et al. (2019)	BiLSTM-CRF	Custom	☑	☑	PE	⊠	⊠
Mulwad et al. (2011)	SVM, Wikitology, BDpedia	Custom	☑	⊠	PE	⊠	⊠
Li et al. (2022b)	Rule-based, IoC recognizer	Custom	☑	⊠	PE	⊠	⊠
Jo et al. (2022)	BERT-BiLSTM-CRF	Custom	☑	☑	PE	⊠	⊠
Li et al. (2022a)	EEMAP-BERT, POS, Neo4j	Custom	☑	☑	PE	☑	⊠
Injy Sarhan (2021)	RoBERTa-BiGRU + Att, RoBERTa-BiGRU-CRF, Neo4j	Custom	☑	☑	PE	☑	☑
Piplai et al. (2020)	CRF, NN Regular expressions	Custom	☑	☑	PE	☑	☑
Proposed models	SecureBERT-BiLSTM-TDD+ Att, BiLSTM-CRF + Att, Domain-ontology, Neo4j	Public	☑	☑	PE	☑	☑

(entity1, relation12, entity2). In this format, entity1 and entity2 denote the nodes (or entity categories), while relation12 signifies the relationship between them.

State of the art of Joint extraction technique

2.2.1. Joint extraction technique

The joint extraction approach integrates named entity recognition (NER) and relation extraction (RE) into a single model to reduce error propagation and enhance accuracy. This model employs multi-task learning, beginning with entity identification and followed by relation extraction. This process produces entity-relation triples that are used to construct the knowledge graph.

Several studies have adopted the joint extraction approach, notably the work of Miwa and Bansal (2016), who proposed a joint relation extraction model that utilizes shared parameters to capture both word sequences and dependency tree substructures. This innovative model employs two stacked bidirectional long short-term memory recurrent neural networks (BiLSTM-RNNs) to facilitate end-to-end relation extraction, effectively integrating entity and relationship identification into a single process. Bekoulis et al. (2018) developed a joint model that incorporates a Conditional Random Field (CRF) layer to simultaneously address entity recognition and relation extraction, conceptualizing these tasks as a multi-headed selection problem. This approach allows for the effective capturing of dependencies between entities and their relations within the text. Yuan et al. (2020) introduced a Relation-based Attention Network (RSAN) that leverages a relation-aware attention mechanism built on a BiLSTM architecture. This model enhances the ability to jointly extract entities and relations by focusing on the interdependencies between them, which improves the overall extraction accuracy. Wang et al. (2020c) developed a document-level extraction framework TRIRECO based on graph convolutional network (GCN) to simultaneously complete relation extraction and coreference resolution. Zheng et al. (2017) proposed a novel tagging scheme using BiLSTM that transforms the joint extraction task into a sequence labeling problem. This innovative approach simplifies the extraction process by treating it as a labeling task, where each token in the sequence is tagged with an entity or relation label, streamlining the extraction of complex relationships between entities.

More recently, the extraction of relation triples has been enhanced by using transformer-based architectures. Guo et al. (2021) introduced CyberRel, a system that jointly extracts entity-relation triples from CTI reports for constructing CKGs. CyberRel utilizes the latest advancements in pre-trained models, specifically BERT, to generate word vectors. It then employs a bidirectional gated recurrent Unit (BiGRU) combined with an attention mechanism to extract features effectively. The final decoding of these features into structured outputs is accomplished using a combination of BiGRU and a CRF. Zuo et al. (2022) proposed an attention-based end-to-end model, BERT-BiLSTM-CRF, which implements a novel tagging scheme to jointly extract entities and their relationships from security articles. Similarly, Ahmed et al. (2024) leveraged a pre-trained model based on RoBERTa to extract relation triples from unstructured text reports. This approach utilizes the advanced capabilities of RoBERTa to parse and identify relationships within textual data efficiently.

The reviewed studies employed a joint technique to simultaneously extract entities and relations from security articles. While this approach integrates the tasks effectively, it complicates the integration of error-control techniques, which are crucial for achieving high accuracy in each subtask. Any modification to improve entity detection accuracy or relation classification accuracy requires breaking down and retraining the entire model in a joint extraction approach. Furthermore, joint techniques, such as those by Zuo et al. (2022), do not account for overlapping relations, which impact the model accuracy due to the complexity of the security language. While this simplifies the model, it also omits important information and context about entities and their interrelations, thereby impacting the quality of the knowledge graph. This paper leverages the flexibility of the traditional pipeline technique to separate NER and RE tasks for improved error control. It introduces a domain-specific ontology to ensure that every extracted pair of entities is accurately connected with the correct relation. Additionally, it employs a cybersecurity-specific pre-trained model that captures the nuances of security language, reducing the likelihood of errors and enhancing the accuracy of entity-relation detections. This approach effectively handles overlapping relations and allows for the efficient construction and assessment of the resulting CKG.

In future work, we plan to implement a joint extraction approach that considers overlapping relations and error control, enabling a more comprehensive comparison between both extraction techniques.

Joint extraction pipeline = named entitiy recognition (NER) + relation extration (RE) at the same time

Pipeline extraction technique = NER + RE achived separatly (2 independant modules)

State of the art for Pipeline extraction technique

2.2.2. Pipeline extraction technique

The pipeline approach separates entity extraction (NER) and relation extraction (RE) into two independent modules. In this method, CTI reports are first processed by a NER module to identify and extract relevant entities. The results are then passed to the RE module, which produces the relation triples.

In recent years, numerous studies have utilized the pipeline technique to extract relation triples from reports. Li et al. (2022b) employed a combination of NLP-based and rule-based methods to extract entities and relationships from security reports. Their approach, which leveraged technique templates based on the MITRE repository (MITRE, 2023), enabled the model to achieve promising results in constructing the Technique Knowledge Graph (TKG). Gasmi et al. (2019) utilized BiLSTM and CRF models to extract cybersecurity entities sequentially and their relationships from texts, focusing on comparing their performance. (Zhao et al., 2020) developed a method for identifying Indicators of Compromise (IOCs) using a multi-granular attention mechanism. They further enhanced the accuracy of their method by constructing a heterogeneous information network to model IOC dependencies. Meanwhile, Jo et al. (2022) utilized a BERT pre-trained model combined with BiLSTM and CRF to develop an API framework called Vulcan, which extracts malware entities and relationships from security articles. The authors demonstrated the model's strengths through practical scenarios such as threat evolution identification and profiling.

These studies differ significantly from our work, as they primarily focus on extracting entities and relationships from security reports without addressing the construction and analysis of CKGs. CKGs are central to our research as they offer unique solutions to enhance the performance of security engineers. The Vulcan model from Jo et al. (2022) supports threat analysis but offers only limited search APIs for basic operations such as threat evolution and profiling. Consequently, any new query operation requires technical skills to build a search API and ensure its functionality. Unlike Vulcan, our framework represents the extracted CTI data in the form of a TiKG or CKG built using the Neo4j graph data science library. This allows for complex queries, enabling security engineers to correlate large volumes of CTI reports, extract threat intelligence, anticipate potential future targets, and design appropriate countermeasures in a timely manner. Additionally, previous works used custom datasets with fewer entity classes and relationships, limiting their reproducibility and generalization. For example, Vulcan was trained on a dataset with 6 entity categories and 7 relationships. In contrast, our study employed a public dataset with 13 entity categories and 15 relationships for training the SecureBERT-based model, which was further validated on two other public datasets to ensure reproducibility and generalization. The extraction of a large number of relations between entities often leads to a high rate of classification errors (Peng et al., 2022; Injy Sarhan, 2021), impacting the quality of the resulting CKG. This paper addresses this issue by coupling the relation inference model with a domain ontology that encodes expert knowledge. This approach ensures high accuracy in relation extraction, even for large-scale datasets. Moreover, existing models based on general-purpose languages were implemented as baselines against our security context-aware models to not only demonstrate the quality of our extraction models but also underscore the overall efficacy and effectiveness of our proposed pipeline within a comprehensive threat intelligence system.

Other studies closely related to our work have made initial contributions to the field of cybersecurity knowledge extraction. For instance, Mulwad et al. (2011) applied an SVM classifier to social media texts to identify information describing potential vulnerabilities. From these descriptions, they used Wikitology along with the exploit taxonomy to extract events and concepts, which were then represented using OWL ontology for graph construction. In contrast to our work, they used weak reasoning algorithms based on SVM and focused only on extracting vulnerability entities from texts. Our paper, however, employs advanced NLP algorithms based on transformers and covers a broad range of entity categories and their relationships from security reports.

Li et al. (2022a) introduced a sophisticated threat intelligence information extraction system called Entity Extraction with Multi-head Attention and Part of Speech (EEMAP). This system comprises four main steps: entity extraction, coreference resolution, relation extraction, and knowledge graph construction. The resulting knowledge graph provides a structured overview of the extracted information, showcasing entities, and their interrelations comprehensively. Injy Sarhan (2021) developed the open cyber threat intelligence knowledge graph framework (Open-CyKG), which automates the extraction of information from unstructured APT reports. They employed an attention-based neural open information extraction (OIE) model and a neural cybersecurity NER to effectively extract entities and generate relation triples. These triples are then used to construct the CKG on Neo4j, enhancing the analysis and accessibility of critical cybersecurity information. The main drawback of this approach is the low precision, recall, and F1-score, which results from the model's attempt to extract all relevant security concepts and their relationships from the reports without predefined entities and relations. This can lead to the extraction of a significant amount of unnecessary information. However, security experts are mostly interested in specific types of information defined as STIX domain and relationship objects (Jordan et al., 2002), which they use for threat modeling, detection, analysis, and more. This paper uses these critical threat objects and relationships, implementing an advanced security context-aware model to enhance precision, recall, and F1-scores. Piplai et al. (2020) introduced a novel system designed to automatically extract cybersecurity entities from After Action Reports (AARs). They used a combination of regular expressions (regex) and conditional random fields (CRF) for entity extraction, and RelExt, based on neural networks for relation extraction. This system not only identifies how pairs of entities are related but also inserts them into a CKG using Unified Cybersecurity Ontology (UCO) 2.0. Once the graph is constructed, the pipeline can populate similar attack information from a given security report. The efficacy of the resulting CKG is assessed using various queries, demonstrating the system's capability to synthesize and organize complex information effectively.

While our research utilizes the pipeline technique similar to these studies, we distinguish our approach by integrating SecureBERT contextualized embeddings to enhance the accuracy of entity recognition from cybersecurity texts and minimize error propagation between NER and RE tasks. In addition, our paper introduces a domain-specific ontology in conjunction with an RE attention model. This dual approach helps control classification errors and aims to achieve very high accuracy in the RE task. The aforementioned PE studies not only used custom datasets, which hinder the reproducibility of these works but also used the same dataset for both entity-relation extraction and graph construction. This approach deviates from practical scenarios where trained models are deployed in different environments. In contrast, this paper utilizes a public dataset to train the models, which are then deployed to extract entity-relation triples from curated CTI reports. Similar to the work of Piplai et al. our research refines the integration and analysis processes within the construction of CKGs. We focus on the practical utility of the generated knowledge graphs, emphasizing their applicability and effectiveness in real-world cybersecurity scenarios. This approach ensures that our CKGs are not only accurate but also directly useful in reducing the burden on SOC (security operation center) professionals by identifying potential threats from a large volume of reports.

Table 1 summarizes relevant works on entity-relation extraction from unstructured reports reviewed in this paper. It highlights the differences between various approaches used in the literature and our proposed models for constructing CKGs.

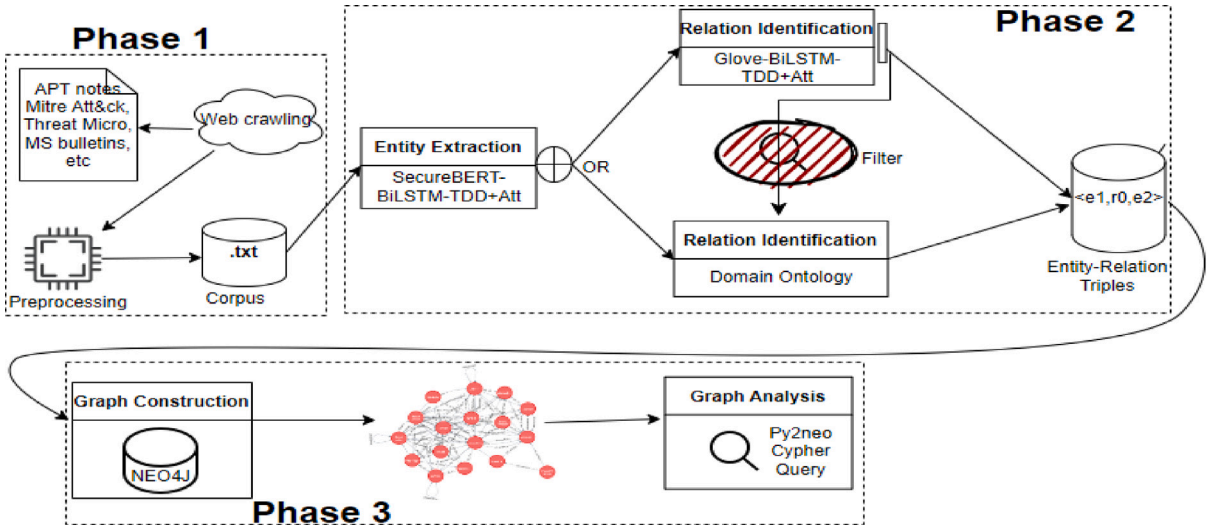

Fig. 1. Threat intelligence knowledge graph framework: TiKG.

3. Threat intelligence knowledge graph framework: TiKG

This section introduces a novel entity-relation extraction pipeline for constructing Cybersecurity Knowledge Graphs (CKGs), known as the Threat Intelligence Knowledge Graph (TiKG) framework, as depicted in Fig. 1. The TiKG framework operates in three main phases:

- **Phase 1: Data Acquisition and Preprocessing** In this initial phase, APT reports are sourced from various authoritative platforms including MITRE ATT&CK Refs. MITRE (2023), the APT Repository (Collections, 2023), Microsoft Threat Center (Global threat activity / microsoft security intelligence, 2005), Trend-Micro (Incorporated, 2023), and many more. These reports are parsed using tools like HTMLParser, PDFLib, and pdftotext. A critical preprocessing step involves cleaning the data by removing non-ASCII characters and rewriting sensitive information into a safer format. For example, email addresses are converted from hacker@email.com to hacker[at]email.com, and URLs from http://www. hacker.com to http://www [.]hacker[.]com. This phase transforms the gathered CTI reports into .txt files, preparing them for the extraction pipeline. It simulates a real-world scenario representing the incoming cyber threat information to the detection system.
- **Phase 2: Entity and Relation Extraction** This phase focuses on extracting entity-relation triples from the processed text reports. It begins with the Entity Extraction module, which employs the most effective entity recognizer model to identify threat entities from the reports. The extracted entities are then passed to the Relation Identification module, which offers two approaches for predicting relations between entity pairs. The first approach utilizes a Domain Ontology that implements a rule-based method to form relation triples. This method is particularly efficient for small to medium datasets. The second approach is a hybrid method that combines the scalability of an AI-trained model with the precision of a domain ontology, and it is ideal for large-scale datasets. The model (e.g., Glove-BiLSTM-TDD + Att) performs initial predictions, and misclassified relations are filtered using the Filter module and sent through the Domain Ontology for error correction. This hybrid method leverages the speed of the AI model during inference and the high accuracy of the ontology to ensure the production of precise relation triples.
- **Phase 3: Construction and Analysis of TiKG** In the final phase, the TiKG is constructed and analyzed. The entity-relation triples generated in Phase 2 are stored in a Neo4j database, where entities serve as nodes and relations as edges. The graph is built using the Neo4j Graph Data Science (GDS) library, which facilitates complex analyses and enables a deeper understanding of the threat landscape.

This structured approach to constructing and analyzing TiKG allows for a comprehensive and systematic analysis of cyber threats, aiding in more informed decision-making and proactive cybersecurity defenses.

3.1. Entity extraction models

It is about finding the best NER model to integrate with our TiKG framework (Fig. 1), that effectively identifies entities within the CTI corpus compared to current state-of-the-art approaches.

3.1.1. NER model

Fig. 2 presents the architecture of our threat entity extraction model, which incorporates SecureBERT and an attention mechanism. SecureBERT, as detailed by Aghaei et al. (2023), is a BERT model pre-trained specifically on extensive cybersecurity-related texts. This pre-training allows SecureBERT to capture domain-specific terminology, jargon, and nuances prevalent in cybersecurity literature, which general-purpose models may not represent as well. By leveraging SecureBERT, our model effectively understands and processes the unique language patterns and contextual information inherent in cybersecurity texts, leading to more accurate entity recognition and relation extraction. The architecture combines these contextualized embeddings with an attention mechanism, a BiLSTM layer, and a Time-Distributed Dense layer (TDD). The TDD layer applies a fully connected (dense) layer to each time step of the input sequence independently. This allows the model to better capture and utilize the features extracted by the BiLSTM layer for each token in the sequence, ensuring that the contextual embeddings provided by SecureBERT are effectively transformed into meaningful features for each time step. This integration significantly minimizes the likelihood of errors propagating through the extraction pipeline and enhances the overall accuracy and effectiveness of entity recognition. The key components of the model presented in Fig. 2 are summarized as follows:

1. Embedding:

$$e_i = E(X_i) \quad \forall i \in \{1, \dots, n\}, \tag{1}$$

where X is the input sequence of length n and X_i is the ith token in the sequence. $E(X)$ as the embedding function applied by SecureBERT, outputting embedded representations e_i for each token X_i.

A voir : détails de fonctionnement du TDD

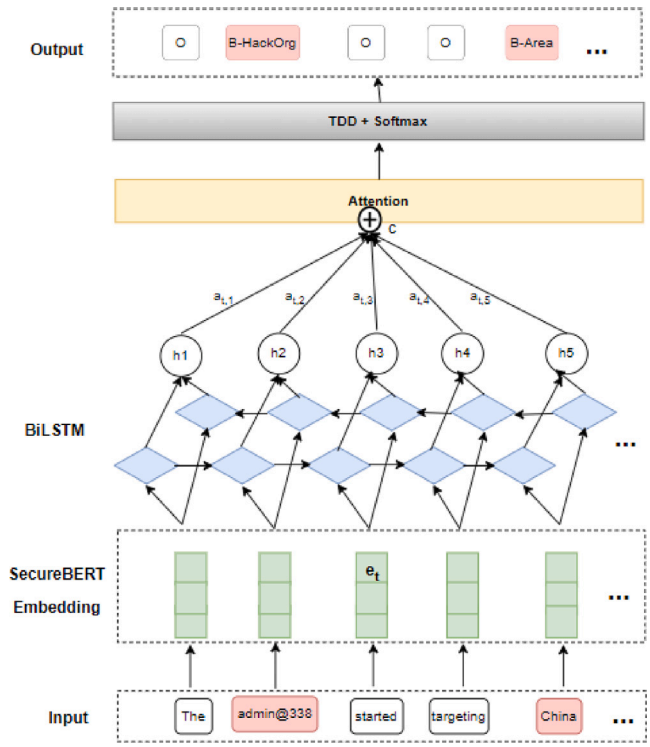

Fig. 2. Entity extraction's model based on SecureBERT.

2. BiLSTM:

$$H = \text{BiLSTM}(e_1, e_2, \dots, e_n) \tag{2}$$

Here, H is a sequence where each h_i represents the hidden state of the BiLSTM for the ith token in the sequence.

3. Attention: If q denotes the mean of all hidden states h_i i.e $q = \frac{1}{n}\sum_{i=1}^{n} h_i$, the attention scores can be computed as:

$$s_i = \text{V}\tanh(\text{W}_1 q + \text{W}_2 h_i), \tag{3}$$

where V, W_1 and W_2 are all trainable parameters. The scores are normalized using the softmax function to obtain the attention weights a_i given by:

$$a_i = \frac{\exp(s_i)}{\sum_{k=1}^{n}\exp(s_k)} \tag{4}$$

4. Concatenation: The context vector c is concatenated with the respective hidden state h_i before passing to the TimeDistributed Dense (TDD) layer.

$$\hat{h}_i = c + h_i \quad \forall i \in \{1, \dots, n\}, \tag{5}$$

where $c = \sum_{i=1}^{n} a_i h_i$.

5. TimeDistributed Dense + Softmax: The concatenated vectors $\hat{h}_i$ are passed through a TDD layer followed by a softmax to predict the token labels.

$$o_i = \text{softmax}(\text{TDD}(\hat{h}_i)) \quad \forall i \in \{1, \dots, n\} \tag{6}$$

The softmax function can be given by:

$$P(Y_i = j|X) = \frac{e^{z_{i,j}}}{\sum_{k=1}^{K} e^{z_{i,k}}}, \tag{7}$$

where $P(Y_i = j|X)$ is the probability that the ith token in sequence X is assigned to the jth label. $z_{i,j} = \text{TDD}(\hat{h}_i)_j$ ie the logits for the jth label and ith token from the TDD layer. K represents the total number of unique labels (or classes)

Table 2
DNRTI dataset information.

Entity types	Meaning	Examples	Count
Area	Location	China	3435
Exp	Exploit	EternalBlue	1995
Features	Characteristics	Lateral movement	2442
HackOrg	Hacker group	MuddyWater	4490
OffAct	Attack pattern	Brute force	2682
Org	Industry	Military industry	4696
Purp	Motivation	DoS	2458
SamFile	File	Worksheets	2341
SecTeam	Security team	FireEye	1920
Time	Time	April 20, 2016	2657
Tool	Tool	Clayslide, LaZagne	4917
Way	Technique	Spear-phishing	2037

6. Output:

$$Y_i = \text{argmax}(o_i) \quad \forall i \in \{1, \dots, n\} \tag{8}$$

Y as the output sequence of labels, where Y_i corresponds to the ith token in the sequence. The function $argmax$ is used to select the highest probability label from the output distribution o_i

Indeed, the performance of an NER model in extracting threat entities is also highly dependent on the quality of its training dataset.

3.1.2. Datasets

We utilize the recently published DNRTI dataset for named entity recognition in threat intelligence, as introduced by Wang et al. (2020a). Our choice is driven by the underexplored potential of this NER dataset in detecting threat entities and their relationships, which is crucial for constructing CKGs from APT reports. In addition, we employ the CyNER (Alam et al., 2022) and STUCCO (Bridges et al., 2014) datasets to further validate our NER models. Unlike previous studies that primarily relied on custom datasets, we emphasize model validation across multiple public datasets, enhancing reproducibility and generalization that significantly benefits security practitioners.

3.1.3. Dataset information and preprocessing

The DNRTI dataset contains 36,808 labeled entities, 175,220 tokens, annotated by experts in threat intelligence using the widely recognized "BIO" format, wherein: **B** = beginning of an entity, **I** = inside an entity, and **O** = outside of an entity. It includes 13 entity categories: HackOrg, OffAct, SamFile, SecTeam, Tool, Time, Purp, Area, Idus, Org, Way, Exp, and Features. During the preprocessing phase, we thoroughly analyzed the dataset, removed non-ASCII characters, and eliminated incomplete sentences. Moreover, we merged the 'Idus' category into 'Org' to streamline the dataset by changing the 'Idus' label to 'Org'. We also implemented a Python function to perform coreference resolution specifically for the HackOrg entity type. The examples and distribution of the entity categories in the DNRTI dataset are detailed and summarized in Table 2.

The other two datasets used for model validation include:

- **STUCCO**: This dataset contains CVE and NVD descriptions spanning 15,192 sentences and 680,764 words. It encompasses 15 entity types: application, cveID, edition, file, function, hardware, method, OS, parameter, programming language, relevant term, update, vendor, and version.
- **CyNER**: This dataset consists of five entity categories for analyzing CTI reports: malware, indicators of compromise, system, organization, and vulnerability. These entity classes are presented in 4372 sentences and 105,163 tokens.

The original papers provide more detailed information about the datasets.

Table 3
Models' parameter settings.

Parameters	SecureBERT	BERT	BiLSTM
Batch size	8	8	16
Dropout	0.2	0.2	0.2
Initial learning rate	5e−5	5e−5	5e−5
Hidden layer size	128 × 2	128 × 2	100 × 2
Optimizer	Adam	Adam	Adam
Embedding size	768	768	300
Number of epochs	3	5	10

3.1.4. Training, evaluation and validation

To evaluate our NER models, the DNRTI dataset was divided into training, validation, and testing sets with a distribution ratio of 75%, 10%, and 15% respectively. The implementation parameters are outlined in Table 3. We utilized an exponential decay function for the learning rate, starting at an initial rate of $5e-5$. Additionally, a dropout rate of 0.2 was employed to prevent overfitting, helping to ensure that our model generalizes well on unseen data.

We use standard metrics including Precision (P), Recall (R), F1-score, and Accuracy (Acc) to evaluate our NER models. These performance metrics allow for a comprehensive assessment of the model's effectiveness in correctly identifying and classifying entities from the text. Additionally, we compare our models' performances against other state-of-the-art NLP algorithms to gauge their relative strengths.

Table 4 demonstrates that models incorporating SecureBERT consistently outperform those using BiLSTM-CRF, BERT, and standalone RoBERTa across all three datasets: DNRTI, CyNER, and STUCCO. The SecureBERT-BiLSTM-Attention-TDD model achieves the highest precision, recall, and F1 score across all datasets, indicating its superior ability to accurately identify relevant entities within diverse CTI corpora. Specifically, it attains a precision of 0.89, recall of 0.91, and F1 score of 0.90 on DNRTI, while also performing exceptionally well on CyNER (P: 0.95, R: 0.95, F1: 0.95) and STUCCO (P: 0.97, R: 0.98, F1: 0.97). These results underscore the model's robustness and adaptability to different datasets. Additionally, the highest accuracy achieved by SecureBERT-BiLSTM-Attention-TDD across the datasets (0.98 on DNRTI, 0.99 on CyNER, and 0.99 on STUCCO) suggests its consistent performance in correctly classifying entities. This superior performance of the SecureBERT models highlights their distinct advantage in entity extraction tasks within the CTI domain. The inclusion of attention mechanisms further enhances their effectiveness, implying the model's adeptness at resolving ambiguities and recognizing context-specific terms and relationships essential for interpreting the specialized language of cybersecurity. This empirical evidence validates the theoretical considerations of using SecureBERT contextual embeddings to reduce error propagation, leading to a more precise classification of tokens into their respective entity types. All models show relatively strong performance on the STUCCO dataset because it contains vulnerability descriptions with well-known patterns that differentiate between entity types. Similarly, CyNER describes APTs but features only five distinct categories. This is not the case with the DNRTI dataset, which describes APT reports and lacks such distinct patterns between entity types. For instance, general-purpose models face challenges with language ambiguity in differentiating between entities such as Features, OffAct, Purp, and Way, which require sentence context for more precise classification. This work used a context-aware model to improve the F1 scores, as shown in Table 4, compared to existing works such as Jo et al. (2022), which used a dataset with a fewer number of entities and relations.

The performance of the SecureBERT-BiLSTM-Att-TTD model is further validated in Fig. 3, which shows that the loss on both training and validation datasets decreases significantly across epochs. This decline in loss indicates that the model is effectively learning from the training data and generalizing well to unseen data in the validation set. Concurrently, there is an observable upward trend in accuracy for both training and validation sets. Starting from an accuracy of 94.42% in training and 97.09% in validation during the first epoch, the accuracy rises to 98.42% and 98.26% respectively by the third epoch.

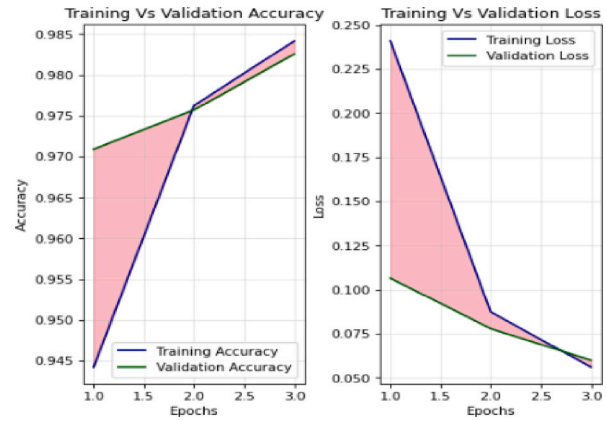

Fig. 3. SecureBERT-BiLSTM-Att-TTD: Best NER model's learning and validation plots.

These observations highlight a robust level of performance, establishing the SecureBERT-BiLSTM-Att-TTD model as the ideal candidate for integration into our TiKG framework (Fig. 1). The demonstrated ability of this model to effectively learn and generalize suggests that it will be highly effective in enhancing the accuracy and reliability of entity recognition within our cybersecurity intelligence system.

3.2. Relation extraction models

This section introduces two relation extraction approaches deployed simultaneously in the TiKG pipeline to extract entity-relation triples from CTI reports. The first approach utilizes a domain ontology, which provides a structured framework of concepts and their relationships specific to the DNRTI dataset. We selected this dataset not only because it contains recent APT reports but also because its potential for threat detection and constructing CKGs remains unexplored. The second approach employs an NLP-based model, which uses natural language processing techniques to dynamically identify and extract relationships between entities directly from text. Together, these methods aim to enhance the comprehensiveness and accuracy of the extracted data, facilitating a more robust and informative TiKG.

3.2.1. DNRTI domain ontology for relation extraction

This method leverages predefined rules and relationships to link entities based on the ontology's schema accurately. We adopted the idea behind the Unified Cyber Ontology (UCO) 2.0 framework from Syed et al. (2016) to define a domain-specific ontology for the DNRTI dataset. This ontology includes 15 specific relations, each meticulously described to capture the intricate connections within cybersecurity data. The relations are structured to facilitate precise and contextually appropriate linkage between entities, enhancing both the depth and the utility of the extracted information in TiKG.

- analyses: Domain:secTeam
  Range:SamFile
- associatedWith: Domain:HackOrg.
  Range: HackOrg
- discovers: Domain:SecTeam
  Range: HackOrg
- discoveredBy: Domain:HackOrg
  Range:SecTeam
- hasAttackTime: Domain:HackOrg or OffAct or Way
  Range:Time

Table 4
NER models: Evaluation results, comparisons and validations across DNRTI, CyNER and STUCCO datasets.

Models	DNRTI				CyNER				STUCCO			
	P	R	F1	Acc	P	R	F1	Acc	P	R	F1	Acc
BiLSTM-CRF	0.67	0.72	0.71	0.92	0.76	0.75	0.76	0.96	0.76	0.77	0.75	0.92
BERT	0.77	0.82	0.79	0.96	0.81	0.82	0.81	0.95	0.84	0.87	0.86	0.97
BERT-BiLSTM-CRF	0.78	0.81	0.79	0.96	0.83	0.82	0.82	0.95	0.96	0.96	0.96	0.99
RoBERTa	0.81	0.85	0.83	0.97	0.82	0.87	0.85	0.97	0.93	0.94	0.93	0.98
RoBERTa-BiLSTM-TDD	0.81	0.85	0.84	0.96	0.93	0.93	0.93	0.98	0.96	0.97	0.96	0.99
RoBERTa-BiLSTM-CRF	0.83	0.86	0.84	0.96	0.93	0.95	0.94	0.98	0.96	0.97	0.96	0.99
SecureBERT-BiLSTM-TDD	0.86	0.88	0.87	0.96	0.95	0.95	0.94	0.99	0.97	0.97	0.98	0.99
SecureBERT-BiLSTM-TDD-Att	**0.89**	**0.91**	**0.90**	**0.98**	**0.96**	**0.96**	**0.96**	**0.99**	**0.98**	**0.98**	**0.98**	**0.99**

- hasCharacteristics: Domain:HackOrg or OffAct or Exp or Way or Tool or SamFile.
  Range:Features
- locatedAt: Domain:Org Range:Area
- monitors: Domain:SecTeam
  Range:Org or Area or Tool or Exp
- monitoredBy: Domain:Org or Area or Tool or Exp
  Range:SecTeam
- motivates: Domain:Purp
  Range:HackOrg or OffAct or Exp or Way
- motivatedBy: Domain:HackOrg or OffAct or Exp or Way.
  Range:Purp
- uses: Domain:HackOrg or OffAct or Exp or Way or Tool or SamFile.
  Range:Tool or OffAct or Exp or SamFile or Way
- usedBy: Domain:Feaures or OffAct or Exp or Way or Tool or SamFile.
  Range:HackOrg or OffAct or Exp or Way or Tool or SamFile
- targets: Domain:HackOrg or OffAct or Exp or Way or Tool or SamFile.
  Range:Area or Org or SecTeam
- targetedBy: Domain:Area or Org or SecTeam
  Range:HackOrg or OffAct or Exp or Way or Tool or SamFile

Fig. 4 presents a visual representation of the sample relations defined by the domain-specific ontology between various security entity categories within the DNRTI dataset. This illustration helps clarify how entities such as hacker organizations, exploits, tools, and many more are interconnected within the ontology framework. This figure serves as a crucial tool for both understanding the structured relationships within the TiKG and for developers and analysts working to integrate or enhance the ontology's application in real-world cybersecurity scenarios.

3.2.2. Relation extraction models

This subsection explores the application of two NLP-based methods for relation extraction from unstructured security articles.

- **Attention-based Bidirectional Long Short-Term Memory (Att-BiLSTM)**: This model, proposed by Zhou et al. (2016), effectively captures semantic information within sentences. It utilizes an attention mechanism that automatically focuses on words that have a decisive impact on classification. This feature allows the model to perform accurately without the need for additional knowledge bases or complex NLP systems.
- **Attention-based Convolutional Neural Networks (Att-CNN)**: Introduced by Shen and Huang (2016), this model combines word embeddings, part-of-speech tag embeddings, and position embeddings. The integration of these elements with an attention mechanism enables the model to identify the most influential parts of a sentence concerning the two entities of interest.

Adopting these advanced attention-based technologies for relation extraction strategically aims to enhance model focus during training on

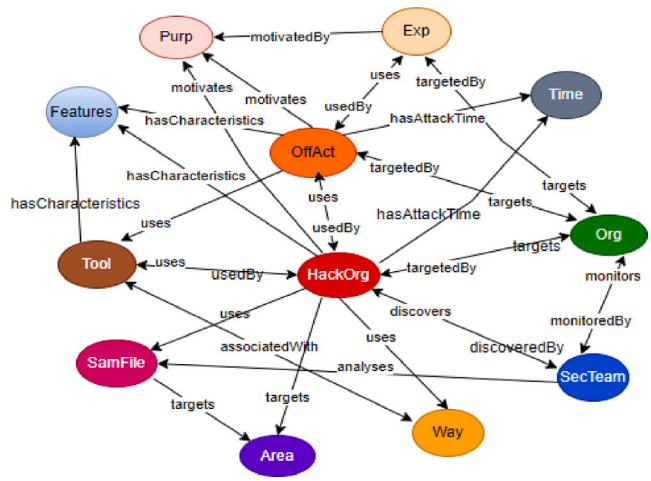

Fig. 4. Sample of the Ontology's schema showing entity relationships.

the threat entities initially identified by the NER model. This targeted focus on critical information improves the accuracy of relation extraction models at inference. By doing so, we address a common challenge in the traditional pipeline technique—maintaining relevance and continuity between the entity recognition and relation extraction modules. This approach effectively mitigates error propagation, enhancing the overall precision and reliability of the system in identifying and linking threat-related entities within complex security texts.

Evaluation and Results: To evaluate the Att-BiLSTM and Att-CNN models on the DNRTI dataset for relation extraction, the dataset needs annotation. We utilized our predefined knowledge ontology, which automatically assigns a relation to every pair of entities in a sentence based on our ontology schema. Entities that do not match any relationships in the schema are discarded. This methodology may result in multiple similar sentences from a single input sentence, each representing a different pair of entities and their defined relationship, potentially increasing the number of sentences proportionally to the number of credible entity pairs identified. Consequently, this elevates the model's processing time complexity.

The annotated DNRTI dataset is then divided into training, validation, and testing sets following a 75:10:15 split. Using the settings outlined in Table 5, we then assess and compare the performances of the Att-BiLSTM and Att-CNN models across different embedding layers, with results detailed in Table 6. This table shows that the Att-BiLSTM-TDD model outperforms the Att-CNN-TDD model in terms of precision, recall, and F1 score, regardless of the embedding used. The GloVe embeddings seem to enhance the performance of both models, with the most notable improvement observed in the Att-BiLSTM-TDD model across all metrics with a precision of 96%, recall of 96%, and F1 score of 96%.

The learning curves shown in Figs. 5 and 6 using glove embeddings, support the conclusion drawn from Table 6 that the Att-BiLSTM-TDD model outperforms the Att-CNN-TDD model.

Table 5
Models' parameter settings.

Parameters	Att-BiLSTM	Att-CNN
Batch size	16	16
Dropout	0.5	0.5
Initial learning rate	1e−5	1e−5
Epsilon	1e−6	1e−6
Weight decay	0.001	0.001
Hidden layer size	100	–
Optimizer	Adam	Adam
Embedding size	300	300
Number of filters	–	100
Window	–	3
pos_dim, pos_dist	–	5, 50
Number of epochs	55	100

Table 6
RE models: Evaluation results and comparisons.

Models	Embedding	Precision	Recall	F1
Att-CNN-TDD	word2vec	0.87	0.88	0.87
	Golve	0.91	0.90	0.91
Att-BiLSTM-TDD	word2vec	0.93	0.92	0.93
	Golve	**0.96**	**0.96**	**0.96**

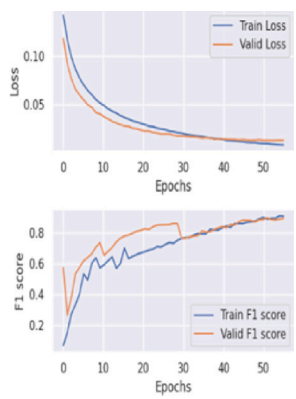

Fig. 5. Att-BiLSTM-TDD.

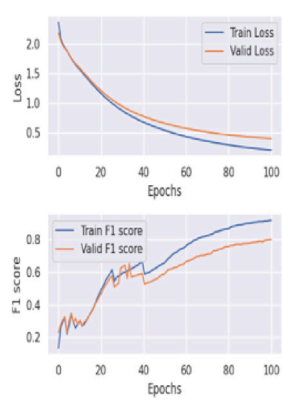

Fig. 6. Att-CNN-TDD.

Loss: Both models show a decreasing trend in training and validation loss over epochs, indicating that they are learning and improving their performance on the DNRTI dataset over time. The Att-BiLSTM model's curves show a steeper decline, suggesting that it is learning more effectively and possibly converging faster than the Att-CNN model, thus requiring less number of epochs.

F1 score: The Att-BiLSTM model demonstrates a steady increase in both training and validation F1 scores as the number of epochs grows, which correlates with its ability to predict the correct labels with high precision and recall consistently as shown in Table 6. The Att-CNN model also improves over time, but the curves suggest a more volatile learning process with fluctuations, especially in the validation F1 score. This indicates that the Att-CNN model is not generalizing as well as the Att-BiLSTM model.

The Att-BiLSTM-TDD model demonstrates more robust and consistent learning, evidenced by both a reduction in loss and an improvement in F1 score. This performance establishes it as the most effective model to be incorporated into our TiKG for relation extraction.

It is important to note the advantage of the domain ontology in the TiKG framework for relation extraction. The domain ontology is leveraged in the preprocessing step to quickly annotate the training dataset for the AI model. In the post-processing stage, the domain ontology acts as a filter, reducing the likelihood of false positives and negatives by verifying the AI model's predictions against predefined rules or expert domain knowledge. This two-fold application ensures higher accuracy in relation extraction.

3.3. Graph construction

This section aligns with Phase 3 of our TiKG framework, as depicted in Fig. 1, where the graph is constructed after extracting entity-relation triples from CTI reports. We deploy the most effective NER and RE models identified from the previous analysis, along with the domain ontology, as specified in Phase 2 of the framework. We then crawl a new set of unstructured security articles from various sources, preprocess them, and direct them to our pipeline framework for entity and relationship extractions. The extracted entity-relation triples are then utilized to construct the CKG.

This approach is designed for a real production environment where models are trained offline on specific datasets (DNRTI in this case) and applied for threat detection on different, yet related, data sets. This method not only addresses the gaps noted in previous works, which often used the same dataset for both training models and building CKGs but also demonstrates the capabilities of the DNRTI dataset in effectively detecting threats and identifying relationships.

3.3.1. CTI dataset

The authors of this paper curated a dataset for constructing the TiKG by crawling various CTI sources as detailed in Section 3, Phase 1. Part of this dataset was provided by Ahmed et al. (2024). Following the preprocessing steps outlined in Section 3, the data was converted into a .txt file format, resulting in a dataset comprising 19,074 tokens and 8037 sentences. This structured dataset forms the basis for further validating the robustness of our proposed TiKG framework.

3.3.2. Graph construction

To construct our threat intelligence knowledge graph or TiKG, we initially process the curated CTI data through the NER-RE pipeline, as shown in Phase 2 of Fig. 1, to extract the entity-relation triples. Within this pipeline, we employ and compare both RE approaches: ontology-based and model + ontology-based to ensure that each path achieves complete accuracy in extracting relationships between entities. This dual-method strategy enables us to validate and refine the extraction process, ensuring the highest level of accuracy and reliability in the relationship mapping within the knowledge graph. This precision is crucial for security professionals who rely on the resulting CKGs for predicting potential future attacks and deploying effective defense strategies.

Night Dragon was a cyber espionage campaign that targeted oil, energy, and petrochemical companies, along with individuals and executives in Kazakhstan, Taiwan, Greece, and the United States.

The text in *Italic* represents a sample sentence from MITRE's CTI reports, which has been processed through our extraction pipeline. The outcomes of the entity recognition are detailed in Table 7, and the

Table 7
Sample sequence tokens and their corresponding predicted tags using SecureBERT-BiLSTM-TTD-Att.

token	⟨s⟩	Night	Dragon	was	a	cyber	espionage	campaign	that	targeted	
pred_tag	O	B-HackOrg	I-HackOrg	O	O	B-OffAct	I-OffAct	I-OffAct	O	O	
token	oil,	energy,	and	petrochemical	companies,	along	with	individuals	and	executives	in
pred_tag	B-Org	B-Org	O	B-Org	I-Org	O	O	O	O	O	O
token	Kazakhstan,	Taiwan,	Greece,	and	the	United	States	⟨s⟩			
pred_tag	B-Area	B-Area	B-Area	O	B-Area	I-Area	I-Area	O			

Table 8
Samples of predicted relations triples.

Entity E1	Relation	Entity E2
Night Dragon	Uses	cyber espionage campaign
Night Dragon	Targets	oil, energy, petrochemical companies
Night Dragon	Targets	Kazakhstan, Taiwan, Greece, the United States
cyber espionage campaign	Targets	oil, energy, petrochemical companies
cyber espionage campaign	Targets	Kazakhstan, Taiwan, Greece, the United States

entity-relation triples derived from this process are reported in Table 8. These tables showcase the accuracy and effectiveness of our models, demonstrating their capability to accurately identify and relate entities within cybersecurity data. Table 8 contains examples of overlapping relations where multiple relationships involve some of the same entities (Night Dragon and cyber espionage campaign) within a single sentence. Our model extracts these detailed relationships that were not considered in a joint extraction technique such as Zuo et al. (2022). In a sentence where entity "cyber espionage campaign" is replaced by the exact name of the malware or ransomware that was used by "Night Dragon", the model makes sure to capture this crucial information to benefit the security professionals.

To construct TiKG, we utilized the desktop version of the Neo4j database management system, which is highly effective for representing and querying interconnected data. We configured a projected graph within Neo4j, where nodes represent the entities and relationships correspond to the relation, both extracted from the CTI corpus by our pipeline. This setup allows us to efficiently manage and analyze the complex network of cybersecurity information captured by our system. The resulting TiKG comprises 6100 nodes representing entities and 31,069 relationships, effectively mapping the complex interactions and connections within the cybersecurity domain. This is depicted in Fig. 7, illustrating the breadth and depth of relationships derived from our CTI data extraction pipeline.

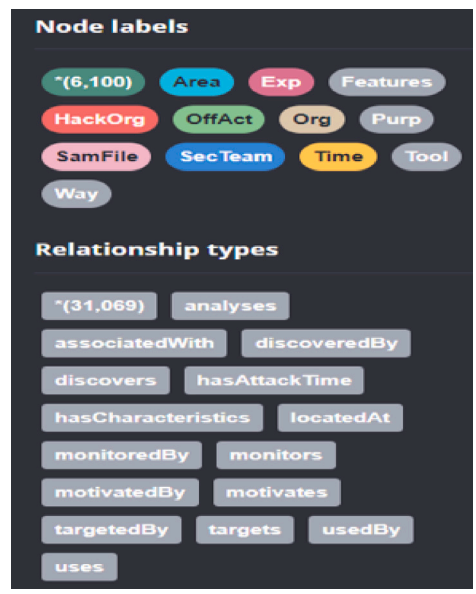

Fig. 7. TiKG: Nodes and relationships.

3.4. Graph analysis via a use-case scenario

The objective of this section is to evaluate the final component of the framework, namely the effectiveness and utility of the constructed TiKG. The current literature lacks a proper framework for evaluating Cybersecurity Knowledge Graphs (CKGs). While Hongyi et al. (2024) proposed a theoretical transformer-based model for this purpose, it overlaps with extraction model experiments (Tables 4 and 6). Consequently, many authors focus solely on extracting entity triples or constructing graphs without further assessment or analysis. Knowing that the main purpose of constructing CKGs is to enhance the workflow and performance of security engineers, our innovative approach utilizes practical cybersecurity scenarios to demonstrate this capability. These scenarios involve correlating historical attack patterns, identifying hacker organizations involved in collaborative attacks, and predicting potential future targets. By leveraging our extraction pipeline, security engineers can quickly access this data through queries, significantly reducing the workload of manually reviewing large volumes of security reports to extract insights about emerging cyber threats. This efficiency improvement ultimately enhances their awareness and security posture. The evaluation of this scenario is structured into two distinct phases:

3.4.1. Phase 1: Identification of collaborative attack relationships

This phase involves detecting or identifying hacker groups (HackOrg) or APTs based on their shared use of common tools (Tool). In the DNRTI dataset, which was used to train our extraction pipeline, the entity category "Tool" refers to software or malware (e.g., Carbanak, BUBBLEWRAP, LaZane, WebShell). These tools are known to incorporate multiple techniques and therefore are very crucial for a successful attack. We define a collaborative attack (CO_ATTACK) as one where hacker groups use common tools in their attack campaigns. Their identification in the CTI corpus is achieved through a query establishing a CO_ATTACK relationship between these groups, signifying their collaboration in attacks. The query also displays the number of resulting nodes and relationships:

```
query1=
"""MATCH (a1:HackOrg)-[:uses]-(:Tool)-[:uses]-(a2:HackOrg)
WHERE a1 <> a2
MERGE (a1)-[:CO_ATTACK]-(a2)
WITH 'CO_ATTACK' AS graphName,
count(DISTINCT a1) + count(DISTINCT a2) AS numNodes,
count(DISTINCT a1)-1 AS numRelationships
RETURN graphName, numNodes, numRelationships """
graph.run(query1)
```

The results of this query are presented in Table 9, which identifies 252 threat groups involved in collaborative attacks, linked by a total

Toute la partie sur les use case scenario est importante

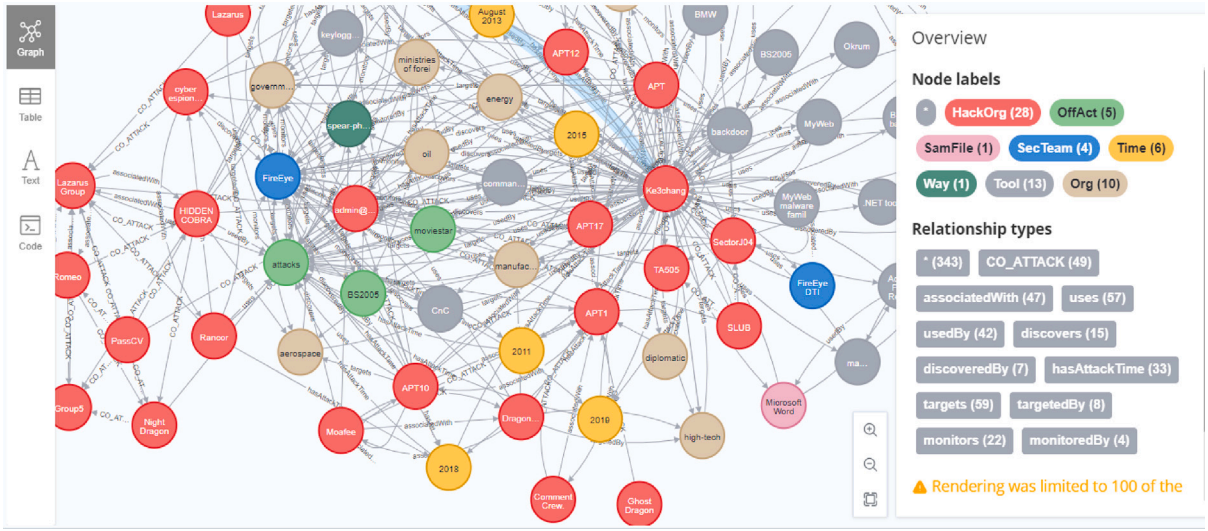

Fig. 8. Sample display of collaborative attack (CO_ATTACK) nodes and relationships.

Table 9
Number of nodes and relations involved in the CO_ATTACK.

graphName	numNodes	numRelationships
CO_ATTACK	252	125

Table 10
Query2: Examples of hacker organizations using common tools.

Entity1	Entity2	CommonTools
PassCV	admin@338	Publicly available RATs
APT1	admin@338	Poison Ivy
EmissaryPanda	APT27	Bitcoin mining malware
APT40	Volt Typhoon	Web shells
LuckyMouse	APT27	Bitcoin mining malware

of 125 relationships. Sample visualization of these entities (nodes) and their relationships (edges) is shown in Fig. 8, rendering limited to only 100.

We display some examples of tools used in the collaborative attacks using the following query

```
query2="""MATCH (Entity1:HackOrg)-[:uses]->(tool:Tool)
<-[:uses]-(Entity2:HackOrg)
WHERE Entity1 <> Entity2
RETURN Entity1, Entity2, COLLECT(DISTINCT tool) AS CommonTools
""" graph.run(query2)
```

The results are presented in Table 10, which shows that hacker groups like EmissaryPanda and APT27 both utilize Bitcoin mining malware in one of their attack campaigns.

3.4.2. Phase 2: Predicting potential future targets

It focuses on predicting potential future targets of hacker organizations leveraging the collaborative attack relationships documented within the TiKG. We employ a specific query designed to identify these potential future targets (represented as a3:Org) of hacker groups (a1:HackOrg) involved in collaborative attacks (CO_ATTACK). This query models a scenario where hacker groups that have previously collaborated in attacks may also target similar organizations in future operations. The query creates MAY_TARGET relationships between the attackers and their predicted future targets, indicating these potential linkages.

```
query3 = """
MATCH (a1:HackOrg)-[r:'CO_ATTACK']-(a2:HackOrg)
MATCH (a2)-[:targets]->(a3:Org)
MATCH (a1)-[:targets]->(a4:Org)
WHERE (a4)<>(a3)
MERGE (a1)-[p:MAY_TARGET]-(a3)
SET p.type = 'MAY_TARGET
""" graph.run(query)
```

We display the results in a DataFrame format for easy reference and analysis using the following piece of code.

```
query4="""" MATCH (a1:HackOrg)-[:MAY_TARGET]->(a2:Org)
RETURN a1.Name as HackOrg, a2.Name as Future_Target
"""" df = pd.DataFrame(graph.query(query4).to_data_frame())
arrays=[df['Potential_Future_Target'],df['HackOrg']]
index = pd.MultiIndex.from_arrays(arrays, names=
('Potential Target', 'Hacker Organizations'))
df = pd.DataFrame(index=index)
display(df)
```

The results reported in Table 11 highlight possible targets of hacker organizations. The link predictions use historical information about the tools or malware that hacker organizations commonly use to infer their potential future targets. Fig. 9 presents visualizations for sample hacker organizations from Table 11.

This use-case scenario generates a detailed report highlighting collaborative attack patterns, potential future targets, and the relationships between hacker organizations, thereby laying the groundwork for enhancing cybersecurity strategies and fortifying defenses against evolving threats. By leveraging practical security scenarios, we provide direct benefits to security practitioners and effectively evaluate the quality and utility of CKGs. We assert that this method offers the most effective means of assessing the efficacy of already populated knowledge graphs.

This also concludes the ablation analysis of our threat intelligence knowledge graph (TiKG) framework, which has shown remarkable performance in extracting security entities and their relationships from various CTI reports. This tool can be extremely useful for security practitioners in predicting future attacks and designing defense strategies.

4. Conclusion and future work

In conclusion, the Threat Intelligence Knowledge Graph (TiKG) framework introduced in this study offers a novel pipeline for extracting threat entities and their relationships from unstructured CTI reports. By integrating SecureBERT embeddings, a domain-specific language model, with an advanced NER-RE pipeline featuring attention-based BiLSTM mechanisms and a domain knowledge ontology, our framework effectively manages the complexities of cybersecurity language and significantly reduces error propagation. This approach not only

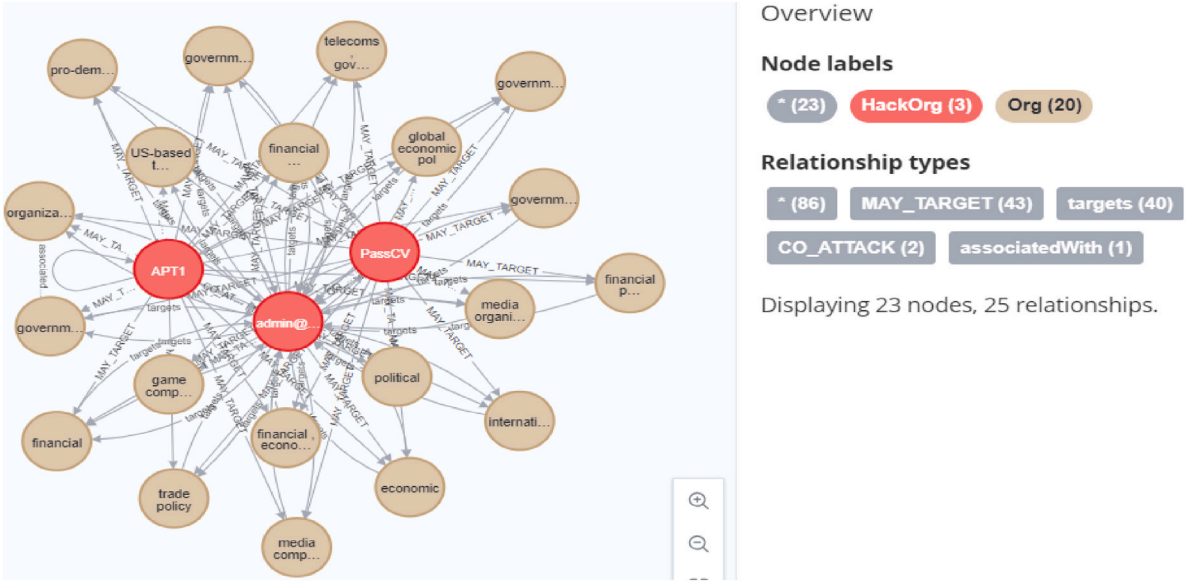

Fig. 9. Visualization of potential targets of some hacker organizations.

Table 11
Potential targets of some hacker organizations.

Hacker organizations	Potential targets
HIDDEN COBRA	Cryptocurrency company, Microsoft customers
Lazarus group	Media industries, finance industries, cryptocurrency, automotive, game companies, etc
APT1, PassCV, admin@33	Financial and policy organizations, media companies, governments, defense sectors, etc
APT10, Moafee	People s Liberation Army, security and news companies, Technology companies, ect
Axiom, Winnti	Government entities, defense industry, law firms, IT companies, chemical manufacturers, etc

enhances the accuracy of entity and relationship extraction but also ensures robustness in identifying nuanced threat patterns within complex textual data. Empirical results show that our TiKG framework outperforms existing systems in constructing more informative and actionable knowledge graphs. The application of the DNRTI dataset further demonstrates the framework's effectiveness in real-world scenarios, providing security professionals with a powerful tool to predict and mitigate potential cyberattacks. As cyber threats evolve, tools like TiKG become increasingly essential for improving the workflow and performance of security engineers. Future work will focus on expanding the framework's adaptability to include more diverse data sources (e.g., user access events, application usage) and refining extraction models to build a full security context-aware framework. Additionally, this study lays the groundwork for a comprehensive comparison of pipeline and joint extraction approaches, and we plan to develop a framework for evaluating CKGs based on practical use cases to enhance threat intelligence, defense strategies, and attack mitigation. This ongoing development aims to continuously improve cybersecurity systems' defensive posture, ensuring they remain effective and resilient against current and future threats.

CRediT authorship contribution statement

Inoussa Mouiche: Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Sherif Saad:** Writing – review & editing, Visualization, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this work are available on our gitHub repository.

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